

Section C

Q: 3(i)

Ans: 3(i)

Cell cycle:-

Cell cycle is the sequence of events involves growth of newly formed cells, replicates its genome and divides into two daughter cells ultimately. It consists of two main phases:-

- * Interphase
- * Mitotic phase.

Interphase :: is the period of cell cycle between two consecutive divisions. It is divided into phases :-

- 1) G_1 (gap₁) phase
- 2) $G_1 S$ (synthesis) phase.
- 3) G_2 (gap₂) phase.

G_1 (gap₁) phase ::

It starts with the production of new cell. cell grows in size, increase the number of its organelles, form needed proteins and other substances. During

cell division, cell prepares itself for the next phase (S) phase. It forms enzymes and nucleotides of DNA required for replication process.

(S) synthesis phase :-

It is the synthesis of DNA. Cell replicates its entire genetic material to form its two copies. That is why chromosomes have 2 chromatids. One for each of the daughter. Each chromatid is attached to one centromere and are called sister chromatids.

G₂ (gap₂) phase :-

During this phase, cell prepares for mitotic phase. Cell forms proteins especially for the formation of spindle fibers. Also cell accumulates energy to complete the cell division. More mitochondria are formed.

Mitotic phase

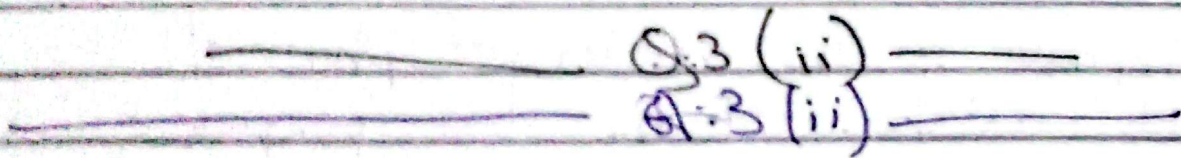
During this phase, cell divides

G₀ phase :-

During G₀, cell may exit ~~and enter~~ ^{the cell cycle}. In G₀ cell stops to divide. Some cell e.g. neurons once mature or never divides again, they remain in G₀ phase ~~forever~~ forever. cell of liver and kidneys etc enter G₀ temporarily. When needed they re-enter the cell cycle and starts to divide again.

Mitotic phase:-

During this phase, cell divides into daughter cell.



Defination:-

The cell division in which parent cell divides into two daughter cells with the same numbers of chromosomes as in the parent cell. ~~It consists~~ but to study easily, it is divided into

- * Karyokinesis
- * Cytokinesis.

Karyokinesis is:-

It is division of nucleus, in which forms two nuclei

in a dividing cell. It is further divided into prophase, ~~anaphase~~ metaphase, anaphase, telophase.

* Prophase :-

During interphase, hereditary material is found in the form of very thin ~~thread~~ thread is called chromatins. If a cell is going to divide, all of its chromatin fibres duplicate during S-phase of interphase.

Prophase is the ~~too~~ longest phase of cell division. At the start ~~prophase~~ prophase chromatin fibres coils and condense into chromosomes. Due to duplication all chromosomes consists of two chromatids. The chromatids of chromosomes are attached to each other at centromere.

Nucleolus disappears as its DNA is packed into chromosomes. At the end of prophase nuclear membrane splits into vesicles which disperse in the cytoplasm. Spindle apparatus is formed.

* Metaphase :-

Chromosomes are aligned to the equator of spindle fibres. Two spindle fibres, one from each pole are attached to the centromere of chromosomes.

* Anaphase :-

Spindle fibres pull the centromeres with which splits and separated chromatids are separated from each other. Individual chromatids are pulled until they reach their respective poles.

* Telophase :-

Chromatids reach at their poles. They uncoil and lengthen to form chromatin fibres again. The spindle apparatus disintegrates. Nuclear membrane is formed around chromatin at each pole. Nucleoli reappear in both nuclei.

Cytokinesis :-

It is division of the cytoplasm which results in the formation of two daughter cells. It starts while telophase is in progress. The pattern of cytokinesis is different in animal and plant cells.

Q: 3 (iii)
A: 3 (iii)

Defination:-

It is a type of cell division which give rise to four daughter cell each having half of the numbers of chromosomes of parent cells. It takes place in cell involded in sexual reproduction. Such parent cells are usually ~~hap~~ diploid which by this reduction division produce haploid daughter cells.

Diploid :-

Cells which have two sets of chromosomes are called diploid ($2n$).

Haploid:-

Cells which have half of the number of chromosomes are called ~~hapo~~ haploid ~~cell~~ (n).

Chromosomes in diploid cells are in homologous pair. Haploid cells carry one member of each homologous ~~cell~~ pair.

Meiosis :

Meiosis is a continuous process but for convenience it is divided into, interphase I, meiosis I, interphase II, meiosis II.

Interphase I :

Before the onset of division process, cells form a copy of its genome by replication process that is why chromosomes appear with two chromatids during division process.

Meiosis I :

It is a reduction process which for daughter cell reduces to the number of chromosomes in half, Firstly, Karyokinesis forms two haploid nuclei which is followed by cytokinesis.

Karyokinesis I in meiosis I :

It is further divided into:-

- 1) Prophase I
- 2) Metaphase I
- 3) Anaphase I
- 4) Telophase I.

Prophase I :-

It is the longest phase of meiosis I. Chromosomes begin to appear as thicker fibres due to coiling of chromatin. Homologous chromosomes line up point to point against each other and form pairs. Their centromeres are in the same position. This pairing process of homologous chromosomes is called synapsis. Each pair is called bivalent.

The non-sister chromatids of homologous chromosomes join each other at some points along their length. Such points are called chiasmata. Each chiasmata is a site for the exchange of segments of non-sister chromatids during prophase I is called crossing over.

To the end of prophase I, chromosomes are fully condensed and paired chromosomes repel each other. However, they are still attached to chiasmata.

Nucleoli and nuclear envelope disappear.

Spindle apparatus is formed.

Metaphase I :-

Bivalents are arranged at the equatorial plate. Spindle fibres are attached to the centromere of chromosomes. One member of a homologous pair receives from one pole and other member from the opposite pole.

Anaphase I :-

Spindle fibres pull on the homologous chromosomes. Separated chromosomes of each pair move to the opposite poles. Each pole receives haploid set of chromosomes.

Telophase I :-

Once at their pole, chromosomes uncoil into chromatin. A nuclear envelope is formed at each pole around the haploid set.

Cytokinesis :-

Cytokinesis occurs by cleavage in animal cell by forming cell wall in plant cell. Although each daughter cell is haploid, their chromosomes are composed of two chromatids. Due to crossing

Over, These chromatids are genetically identical. They must be separated in ~~second~~ second meiosis.

Two haploid daughter cell thus formed enter in interphase II.

————— $Q:3(iv)$ —————
 ————— $A:3(iv)$ —————

Mitosis

Meiosis

Location

It occurs in somatic cells

It occurs in germ line cells.

Division

Cell divides only once.

Cell divides twice.

Daughter cells

It produces two daughter cell.

Four daughter cell are produced.

Haploid / Diploid

Daughter cell produced are diploid.

Daughter cell produced are haploid.

Crossing over

There is no crossing

Crossing over takes

over during mitosis process.

place over meiosis I.

formation of chiasmata

No chiasmata formation

Chiasmata are formed.

Section - B

$G_1: 2(i)$

$A: 2(i)$

Cell cycle: is a sequence of events involved in growth of newly formed cells, replicate its genome and divides it into two daughter cells ultimately. It consists of two main phases.

Interphase:

is a period of cell cycle between two consecutive divisions. It is further divided into G_1 , S, G_2 phase.

Mitotic phase:

During this phase, cells divide into daughter cells.

Q: 2(ii)

A: 2(ii)

G₁:-

It starts with the production of new cells. Cell grows in size. increase the number of its organelles. form the forms needed proteins ~~especially~~ and other substances.

S phase:-

It is the synthesis phase of DNA. Cell replicates its genetic entire genetic material to form its two copies. That is why chromosomes has 2 chromatids. One for each of the daughter cell. Each chromatids are attached to the centromere called sister chromatids.

G₂:-

During this phase, cell prepares for mitotic phase. It form proteins especially for the formation of spindle fibres. Also cell accumulates its energy.

to complete the cell division. More mitochondria are formed.

Q: 2(iii)(or)

A: 2(iii)(or)

Metaphase :-

Chromosomes are aligned to the equator of spindles. Two spindle fibres, one from each pole are attached ~~to~~ at the centromere of chromosomes.

Anaphase :-

Spindle fibres pull the chromatomes and split and chromatids ~~are~~ will be separated from each other, individual chromatids are pulled ~~at~~ until they reach their respective poles.

Spindle fibres exert a pulling force ~~at~~ on the centromeres - ensuring each pole receives an identical set of chromosomes.

Q.2(iv)

A.2(iv)

Telophase ::

Chromatids reached at their poles. They uncoil and lengthen to form chromatin fibres. The spindle apparatus disintegrates. Nuclear membrane is formed around chromatin at each pole. Nucleoli reappears in both nuclei.

Q.2(v)(or)

A.2(v)(or)

Regeneration :-

Some animals have the ability to regenerate the whole part of the body if it is removed accidentally. This is called regeneration.

For example ::

- 1) Sea star - can regenerate its arm.
- 2) Earth worm - can regenerate its head.
- 3) Salamander - can regenerate its limb.

Q:2(vi)

A:2(vi)

Cancer is a disease of cell cycle. Unlike all normal cell of the body. Cancer cell do not have proper functioning cell cycle control system and therefore divides excessively. This excessive growth can result in an abnormal mass of cell called tumour. Not all tumour are cancerous. A benign tumour is an abnormal mass of normal cells. They always remain at their original site in the body. A malignant tumour is cancerous. It is capable of spreading cancer all over neighbouring tissues and often to other distant parts of the body. The spread of cancer cell beyond their original site is called metastasis.

Q:2(vii)

A:2(vii)

Synapsis is the pairing of homologous chromosomes during prophase I. with centromeres in the same position.

Each synapsed pair is called bivalent.

Non sister ~~parent~~ chromatids join at the point called chiasma.

Crossing over is the exchange of segments of genetic material between non-sister chromatids at a chiasma, resulting in genetic recombination.

Q: 2(viii)(or)

A: 2(viii)(or)

Telophase I :

Once at their pole, chromosomes uncoil into chromatin, a nuclear envelope is formed at each pole around the haploid set.

Cytokinesis:-

occurs by cleavage in animal cell by forming a cell wall in plant cell. Although each daughter cell is haploid with half the number of chromosomes, their chromosomes are composed of 2 chromatids. Due to crossing over, these chromatids are not genetically identical. They must be separated in second meiosis I.

$Q: 2(n)$

$A: 2(n)$

Interphase II :-

varies in length but has no further DNA replication, chromosomes are already duplicated from S-phase. Replication

Meiosis II :-

Resembles mitosis; It's divided into prophase II, metaphase II, anaphase II, telophase II.

$Q: 2(n)$

$A: 2(n)$

Humans have 46 chromosomes in their somatic cells. The haploid gametes (egg and sperm) formed by meiosis have 23 chromosomes. In fertilization process a 23 chromosomes sperm fuses with having an egg having 23 chromosomes. The original number of chromosome is 46 is restored in the zygote.

_____ $G: 2(n)$ _____
_____ $A: 2(n)$ _____

Mitosis

1) It occurs in somatic cells

2) It produces two daughter cell

3) No chiasmata is formed

Meiosis

It occurs in germline cell.

~~During~~ four daughter cells are produced.

Chiasmata is formed.

INH

MCQs:

- 1) C
- 2) ~~A~~ B
- 3) C
- 4) D
- 5) B
- 6) C
- 7) B
- 8) C
- 9) D
- 10) B
- 11) B
- 12) C